

3+ Sample testing with continuous outcomes: ANOVA

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Conditions for using ANOVA

Example: Sleep and insight

Example: Cannabis to treat
brain cancer

After a significant
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Bonferroni

Tukey's HSD

Skipping around

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To stick with continuous outcomes, we are going to skip ahead to chapter 24 (ANOVA) and chapter 23 (regression) and bring in some outside information about non-parametric testing.

Summary of continuous outcomes so far - Z and the flavors of t

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In R, the `t.test()` function will run any of the t-tests we have covered so far.

- ▶ One-sample t test, comparing a sample mean to a hypothesized null:
`t.test(data, mu = value)` (if we know σ and have a large sample we could also consider a Z test)
- ▶ Two-sample t test, comparing samples from two independent populations:
`t.test(continuous_var ~ categorical_var, data = df)` or
`t.test(data1, data2)`
- ▶ Paired t test, comparing two non-independent measurements on the same units: `t.test(data1, data2, paired = TRUE)`

Note: the formula interface does **not** accept `paired = TRUE`. For a paired test, pass the two measurement vectors.

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Three sample testing

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Now that we have looked at how to compare one sample to a value and means between two samples, let's extend this to a case where we have 3 samples or 3 groups that we are interested in comparing.

Statistics is everywhere

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PAGING DR. HAMBLIN

Paging Dr. Hamblin: Does Vitamin D Help Fight COVID-19?

Evidence is slowly mounting. Some researchers think that's enough to recommend it.

JAMES HAMBLIN JUNE 3, 2020



Atlantic article

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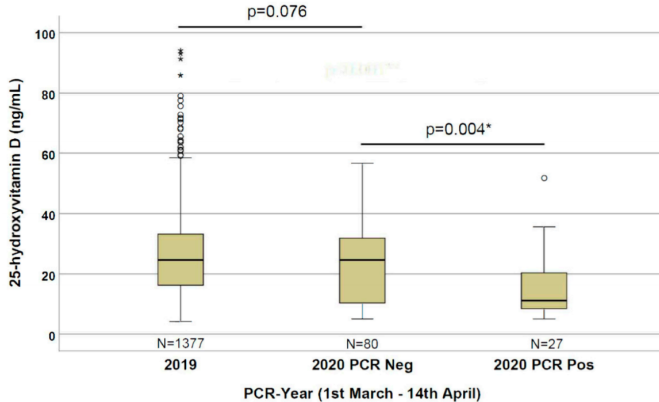


Figure 1. 25-hydroxyvitamin D concentrations in the three evaluated groups (patients from 1 March to 14 April of 2019 and 2020 with a negative PCR, and of 2020 with a positive PCR for SARS-CoV-2.
*: significant results.

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from D'Avolio et al. Nutrients 2020, 12, 1359; doi:10.3390/nu12051359

Today's lecture

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- ▶ Introducing ANOVA - what is the null hypothesis of this test?
- ▶ Visualizing 3 sample data
- ▶ Using ANOVA to test for a difference
- ▶ Example

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ANOVA or analysis of variance can be thought of as an extension of the two-sample t-test to three or more samples.

If we had 3 groups to compare, we could do this by using the two-sample t test multiple times. This would result in $\binom{3}{2} = 3$ comparisons:

- ▶ μ_1 vs. μ_2
- ▶ μ_2 vs. μ_3
- ▶ μ_1 vs. μ_3

The problem with that approach is that we would end up with 3 separate p-values, one for each test performed. None of them tells us how surprising it is to see three sample means spread as far apart as ours are. With more than 3 groups the problem gets worse, because the number of comparisons grows quickly.

We need a single test that gives us one overall measure of evidence. This is a common problem of multiple comparisons.

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Here we are testing a null hypothesis that all the means are the same, for 3 samples this would be

$$H_0: \mu_1 = \mu_2 = \mu_3$$

Our alternative hypothesis is that **at least one** of the means is not equal to the others

Even though your hypothesis involves means, the test compares the variability between groups to the variability within groups

Analysis of variance (ANOVA) is also referred to as the F test.

The ANOVA is based on two kinds of variability:

- ▶ The variability among sample means or how much the individual group means vary around the overall mean
- ▶ The variability within groups, how much do individual observation values vary around the group mean

If the variability within the k different populations is small relative to the variability among their respective means, this suggests that the population means are in fact different.

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Loosely expressed:

$$F = \frac{\text{variation among sample means}}{\text{variation among observations in the same sample}}$$

Note: You will **NOT** need to know the full formula for the F-test, you **will** need to know how to do one in R

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What would the data look like in a data frame?

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What would the data look like in a data frame?

- ▶ One “grouping” variable (categorical)
- ▶ One continuous response variable

ANOVA asks if there is an association between the grouping variable and the response variable.

Visualizing first

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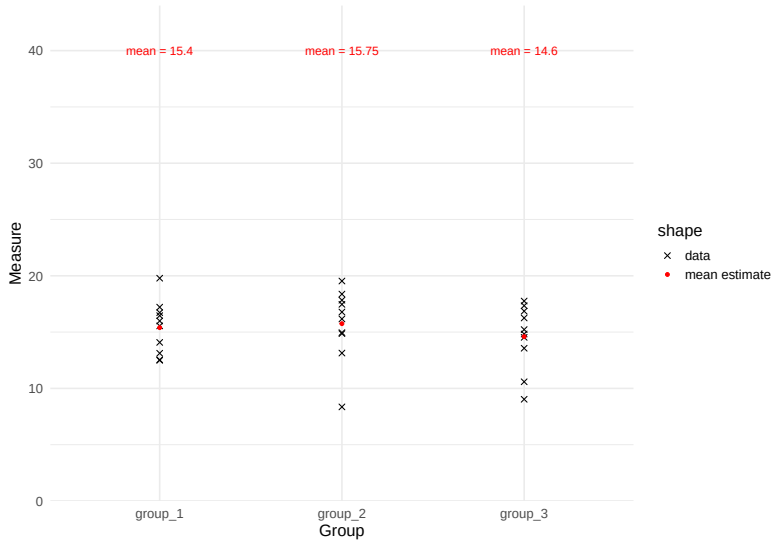
Bonferroni

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What graphical strategies have we learned to look at variability within and between groups?

A) Is there a difference between these means?

Describe why you do or do not think so.



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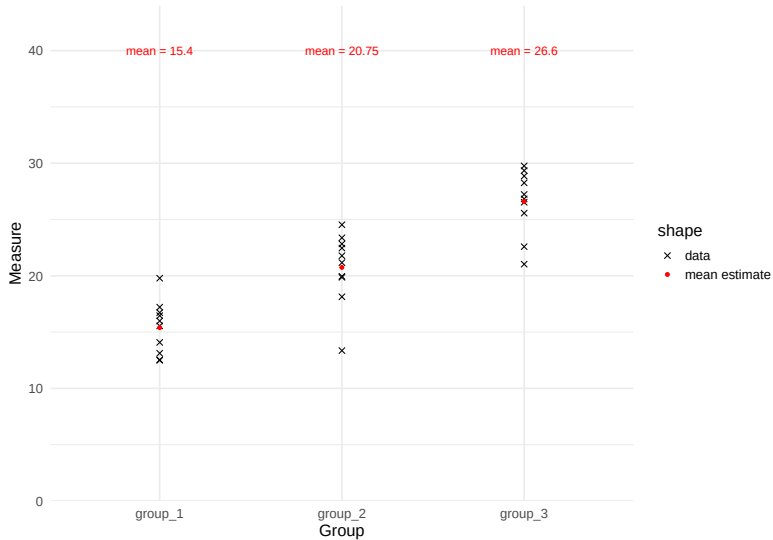
Tukey's HSD

Plot (A)

- ▶ The means (red dots) were not very different across the groups. This means the variation **between** the group means was small.
- ▶ The distribution of the data (black Xs) was wide enough that the distribution of points for each group overlapped almost completely. This means that the variation **within** each group was relatively wide

B) Is there a difference between these means?

Describe why you do or do not think so.



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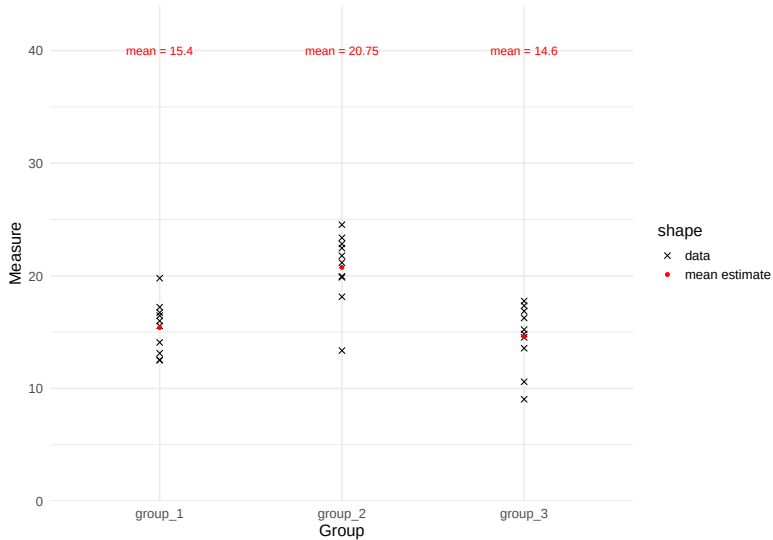
Tukey's HSD

Plot (B)

- ▶ The means are quite different across the groups. The variation **between** the group means would be larger than in plot (A)
- ▶ The distribution of the data overlaps between groups 1 and 2 and 2 and 3, but not 1 and 3. The variation **within** each group is as wide as it was in Plot (A) but doesn't mask the mean differences, especially between group 1 and 3

C) Is there a difference between these means?

Describe why you do or do not think so.



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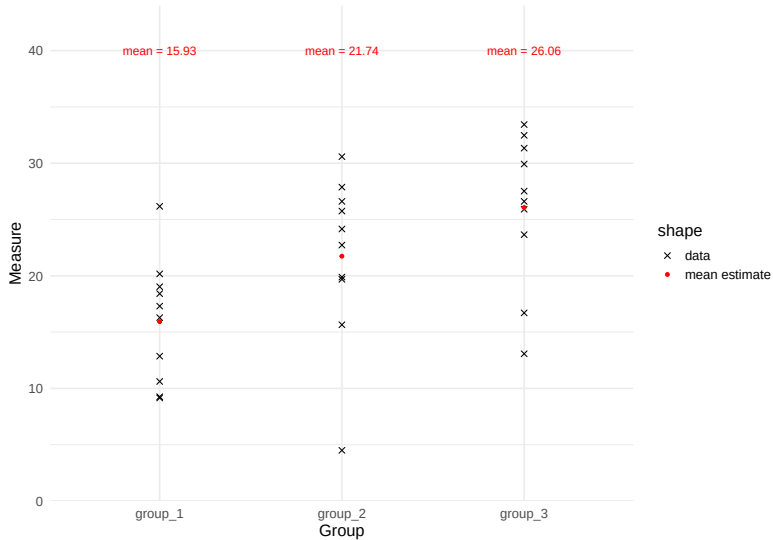
Tukey's HSD

Plot (C)

- ▶ Here, the means for group 1 and 3 look similar, but the mean for group 2 appears a bit higher than the other two, though there is still overlap between the data from all the groups
- ▶ Is there evidence that at least one of the means is different?

D) Is there a difference between these means?

Describe why you do or do not think so.



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Plot (D)

- ▶ Plot (D) is Plot (B) generated again from the same population means, but with the within-group standard deviation raised from 3 to 7
- ▶ That extra variation **within** the groups makes the same difference between the means much harder to detect

Overall summary

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- ▶ What we informally did on the previous slides was compare the variation **between** group means to the variation **within** the groups
- ▶ This focus on variation is why this test is called ANOVA: an ANalysis Of VAriance
- ▶ When the ratio of between vs. within variation is large enough then we detect a difference between the groups
- ▶ When the ratio isn't large enough we don't detect the difference.
- ▶ This ratio is our test statistic, denoted by F

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What other ways to present the data visually have we learned that might be useful before we move on to testing?

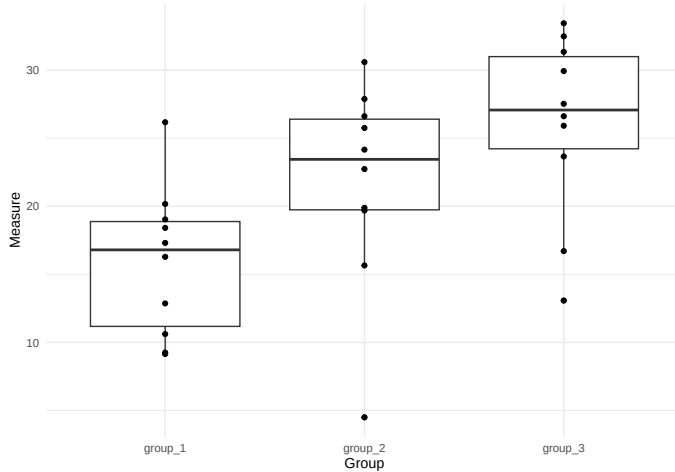
Descriptive plots

How would you want to plot these data before you conduct a test?

- Option 1: Box plot for each level of the grouping variable (with overlaid data points)

```
ggplot(diff_3_narrow, aes(x = Group, y = Measure)) +  
geom_boxplot() +  
geom_point() +  
theme_minimal(base_size = 15)
```

Box plot



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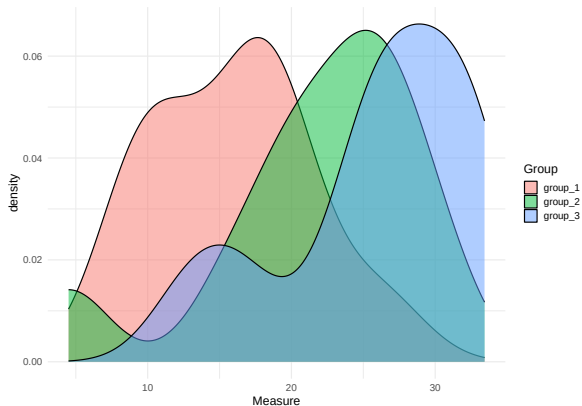
Descriptive plots

How would you want to plot these data before you conduct a test?

- Option 2: Density plot for each level of the grouping variable

```
ggplot(diff_3_narrow, aes(x = Measure)) +  
  geom_density(aes(fill = Group), alpha = 0.5) +  
  theme_minimal(base_size = 15)
```

Density plot



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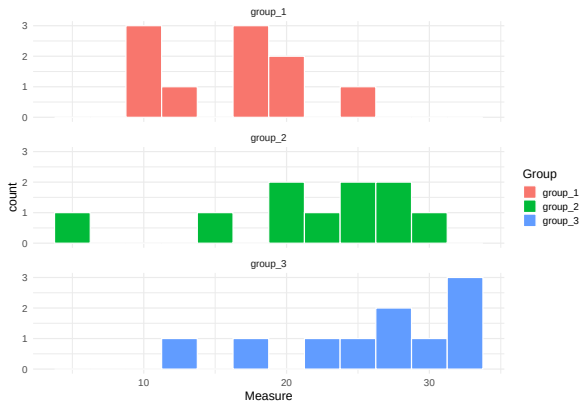
Tukey's HSD

How would you want to plot these data before you conduct a test?

- Option 3: Histogram for each level of the grouping variable

```
ggplot(diff_3_narrow, aes(x = Measure)) +  
  geom_histogram(aes(fill = Group), col = "white", binwidth = 2.5) +  
  theme_minimal(base_size = 15) +  
  facet_wrap(~ Group, nrow = 3)
```

Histograms with facet wrap



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Testing with ANOVA

The hypotheses

Null hypothesis

$H_0 : \mu_1 = \mu_2 = \dots \mu_K$, where K is the number of levels of the grouping variable

- Can you also state the null hypothesis in words?

The hypotheses

Alternative hypothesis

H_a : not all $\mu_1, \mu_2, \dots, \mu_K$ are equal

- In words: Not all means are the same. Or, **at least one of the means differs** from the others.



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The test statistic (ANOVA F Statistic)

$$F = \frac{\text{variation among group means}}{\text{variation among individuals in the same group}}$$

- ▶ Numerator is, fundamentally, the variance of the sample means
- ▶ Denominator is, fundamentally, an average of the group variances.
- ▶ The F statistic follows an F distribution
- ▶ Computation details are at the end of the book chapter (these computation details will not be tested)

The F distribution

- ▶ Skewed right
- ▶ Take only positive values
- ▶ The F distribution depends on the number of means being compared and the sample size for each of the groups
- ▶ Let k be the number of groups being compared and $N_{Total} = n_1 + n_2 + \dots + n_k$ (the total sample size across all the groups)
- ▶ Then the F statistic follows an F distribution with $k - 1$ degrees of freedom in the numerator and $N_{Total} - k$ degrees of freedom in the denominator
- ▶ The p-value of the ANOVA F statistic is always the area to the right of the test statistic

ANOVA in R: use `aov()`, then `tidy()` it up!

- ▶ `aov()` stands for analysis of variance

The general syntax for the ANOVA is:

```
aov(outcomevariable ~ groupvariable, data = dataset)
```

We will save the output of this as an object and then use `tidy(object)` to get the output we want.

reference: https://broom.tidyverse.org/reference/anova_tidiers.html

ANOVA in R: use `aov()`, then `tidy()` it up!

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We will focus on two parts of the output from this package:

- ▶ `statistic` is the F test statistic, the ratio of the variation between means vs. the variation within groups.
- ▶ `p.value` is the p-value for the test.

p of an f statistic in R

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You can check that you can calculate the p-value from the F distribution.
Remember, that you need to specify a degrees of freedom for the numerator and
for the denominator:

`pf(value, df1=numerator degrees of freedom, df2= denominator degrees of
freedom, lower.tail=F)`

This general pattern of syntax should look familiar by now. . . .

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Condition 1: k independent SRSs, one from each of k populations.

- ▶ The most important assumption, because this method, like the others in Part III of the course, depends on the premise of having taken a random sample.

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Condition 2: Each of the k populations has a Normal distribution with an unknown mean μ_i .

- ▶ This assumption is less necessary
- ▶ The ANOVA test is **robust** to non-Normality.
- ▶ Remember that the ANOVA is based on comparing the differences of sample means
- ▶ What did the CLT tell us about variability of sample means when the samples are not normally distributed?

Conditions for ANOVA

Condition 3: All the populations have the same standard deviation σ , whose value is unknown.

- ▶ Hardest condition to satisfy and check
- ▶ If this condition is not satisfied ANOVA is often okay if the sample sizes are large enough and if they are similar across the groups
- ▶ Can use `group_by()` and `summarize()` to calculate the sample SDs to see if they're similar and indicative that the population parameters are too
- ▶ General rule: we want the largest sample standard deviation to be less than twice as large as the smallest one. I.e., $s_{max}/s_{min} < 2$

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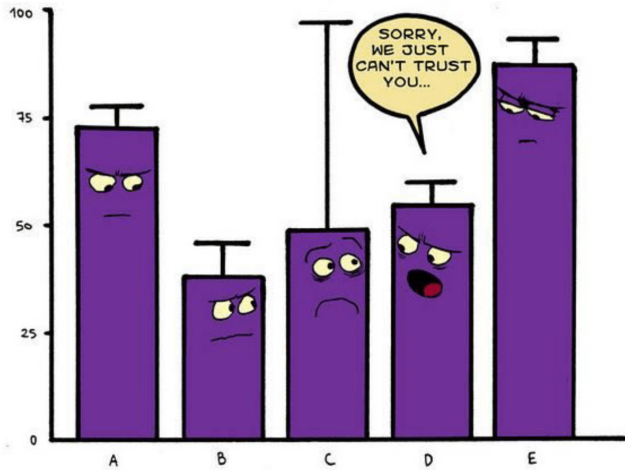
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Example: Sleep and insight

From a 2004 article in Nature

Sleep inspires insight

Ullrich Wagner¹, Steffen Gais¹, Hilde Haider², Rolf Verleger³ & Jan Born¹

¹*Department of Neuroendocrinology, University of Lübeck, Ratzeburger Allee 160, 23538 Lübeck, Germany*

²*Institute of Psychology, University of Cologne, Gronewaldstrasse 2, 50931 Cologne, Germany*

³*Department of Neurology, University of Lübeck, Ratzeburger Allee 160, 23538 Lübeck, Germany*

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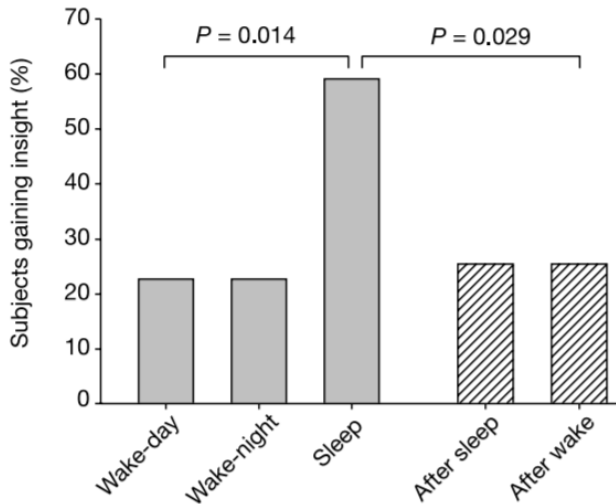
Example: Cannabis to treat brain cancer

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Example: Cannabis to treat brain cancer

Example: Cannabis to treat brain cancer (in mice)

High-grade glioma is an aggressive type of brain cancer with a low long-term survival rate. Cannabinoids, which are chemical compounds found in cannabis, are thought to inhibit glioma cell growth. Researchers transplanted glioma cells in otherwise-healthy mice, and then randomly assigned these mice to 4 cancer treatments: irradiation alone, cannabinoids alone, irradiation combined with cannabinoids, or no cancer treatment. The treatments were administered for 21 days, after which the glioma tumor volume (in cubic millimeters) was assessed in each mouse using brain imaging.

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The data

```
head(cancer_data)
```

```
##      treatment tumor_volume
## 1  Irradiation           30
## 2  Irradiation           46
## 3  Irradiation           46
## 4  Irradiation           95
## 5 Cannabinoids           12
## 6 Cannabinoids           14
```

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Organize the data

- ▶ Think about how you want the data to look.
- ▶ I want to plot the raw data points and display the mean for each treatment group
- ▶ I also want to specify the order that the treatment groups show up in the graph

```
# specify the order of the treatment groups for plotting  
library(forcats)  
cancer_data <- cancer_data %>%  
  mutate(trt_order = fct_relevel(treatment, c("Neither", "Irradiation",  
                                              "Cannabinoids", "Both")))
```

Look at summary statistics

```
# calculate the means and SD for each group
summary_stats <- cancer_data %>%
  group_by(trt_order) %>%
  summarise(mean_vol = mean(tumor_volume),
            sd_vol = sd(tumor_volume),
            samp_size = n())
```

```
summary_stats
```

```
## # A tibble: 4 x 4
##   trt_order    mean_vol sd_vol samp_size
##   <fct>        <dbl>  <dbl>     <int>
## 1 Neither      48.3    24.8         7
## 2 Irradiation  54.2    28.2         4
## 3 Cannabinoids  26      16.6         5
## 4 Both         6       2.76         6
```

Do the conditions hold here?

```
summary_stats %>%  
  summarise(s_max = max(sd_vol), s_min = min(sd_vol),  
            ratio = max(sd_vol) / min(sd_vol))
```

```
## # A tibble: 1 x 3  
##   s_max s_min ratio  
##   <dbl> <dbl> <dbl>  
## 1  28.2  2.76  10.2
```

- ▶ Condition 1 holds by design. The mice were **randomly assigned** to the four treatments
- ▶ Condition 2 is hard to judge with 4 to 7 mice per group, but ANOVA is robust to non-Normality
- ▶ Condition 3 is clearly **violated**. The largest SD is about 10 times the smallest, well above our rule of thumb of 2

What do we do about a violated condition?

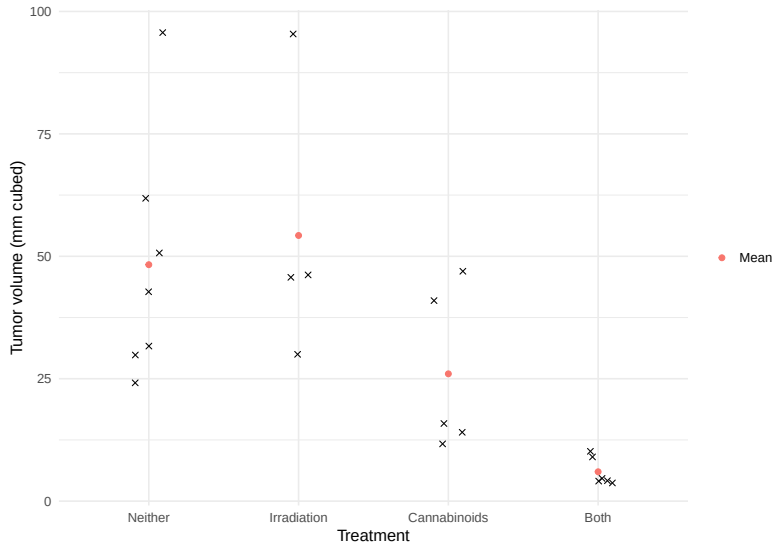
- ▶ The group with the smallest SD is also the group with the smallest mean. When a treatment shrinks tumors toward zero it also shrinks the spread. This is very common for measurements that cannot go below zero
- ▶ We will still run the ANOVA, because it is the tool this chapter teaches, but we should report the result **with that caveat attached**
- ▶ Keep this example in mind. In L21 we will meet non-parametric tests, which do not require equal spreads or Normality

Graph the data

```
ggplot(cancer_data, aes(x = trt_order, y = tumor_volume)) +  
  geom_jitter(pch = 4, width = 0.1) +  
  geom_point(data = summary_stats, aes(y = mean_vol, col = "Mean"), pch =  
    19) +  
  labs(y = "Tumor volume (mm cubed)", x = "Treatment", col = NULL) +  
  theme_minimal(base_size = 15)
```

note: `geom_jitter()` with `width = 0.1` randomly “jitters” the location of the points along the x axis so that we can see each of them since some have the exact same values.

Graph the data



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ANOVA in R: use aov(), then tidy() it up!

```
library(broom)
cancer_anova <- aov(tumor_volume ~ treatment, data = cancer_data)
tidy(cancer_anova)
```

```
## # A tibble: 2 x 6
##   term          df sumsq meansq statistic  p.value
##   <chr>      <dbl> <dbl>   <dbl>      <dbl>    <dbl>
## 1 treatment      3 8060.   2687.        6.70  0.00313
## 2 Residuals    18 7218.    401.         NA      NA
```

This F says that the variation between the means is nearly 7 times as large as the variation within the groups.

This p-value is equal to 0.3%. There is a 0.3% chance of observing the F statistic we observed (or more extreme) under the null hypothesis that all the means are the same.

P value of the F test

Remember

- ▶ k is the number of groups being compared and $N_{Total} = n_1 + n_2 + \dots + n_k$ is the total sample size across all the groups.
- ▶ The F statistic follows an F distribution with $k - 1$ degrees of freedom in the numerator and $N_{Total} - k$ degrees of freedom in the denominator
- ▶ The p-value of the ANOVA F statistic is always the area to the right of the test statistic

```
pf(6.699489, df1 = 3, df2 = 22 - 4, lower.tail = F)
```

```
## [1] 0.003131703
```

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- ▶ The p-value equaled 0.003, indicating a difference. But what groups are actually different?

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After a significant ANOVA...

- ▶ You could look at all pairwise differences (i.e., comparing each combination of treatments), but we have to be careful because we will find differences “just by chance” if we compare enough groups.

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A key example of what not to do?

FOOD FOR THOUGHT

Cornell Food Researcher's Downfall Raises Larger Questions For Science

September 26, 2018 · 3:07 PM ET

BRETT DAHLBERG

FROM WXXI



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A brief reminder about p-hacking

Remember, one of the issues with multiple comparisons is that when you repeatedly question the same dataset, you can end up finding “significant” results by chance alone.

We talked about this before as p-hacking or p-fishing or data dredging

This along with other issues that are sometimes unconscious can lead to bias in what is found and what is published.

Ioannidis, John P.A. (August 30, 2005). “Why Most Published Research Findings Are False”. PLoS Medicine. <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0020124>

Strategies for multiple comparisons

- ▶ Conduct multiple pairwise tests and adjust the critical value for significance using a Bonferroni correction
- ▶ Use a Tukey's HSD to generate the pairwise comparisons with adjusted p-values

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Bonferroni Adjustment for multiple comparisons

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We could do a series, comparing each combination of groups in pairs or $\binom{k}{2}$ comparisons.

To compensate for making multiple comparisons and set the overall probability of making a type I error at 0.05, we can adjust our α to α^* for each comparison by dividing by the number of comparisons we are making.

$$\alpha^* = \left(\frac{0.05}{\binom{k}{2}} \right)$$

We then use α^* as the significance level for each individual comparison. So for a comparison of 3 groups we would use an α of 0.0167 as the significance level for each comparison.

This modification is known as the Bonferroni correction. Bonferroni is fairly basic and can become unwieldy - what happens if you have a lot of groups?

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Bonferroni for our example

This would entail doing a series of two-sample t tests for each pairwise comparison or 4 choose 2 comparisons = 6 two-sample tests so the adjustment would be:

$$\alpha^* = \left(\frac{0.05}{\binom{4}{2}} \right) = \left(\frac{0.05}{6} \right) = 0.0083$$

Bonferroni for our example

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So we would pull out each group and conduct each pairwise test:

```
both <- cancer_data %>% filter(treatment == "Both") %>%  
  pull(tumor_volume)  
neither <- cancer_data %>% filter(treatment == "Neither") %>%  
  pull(tumor_volume)  
can <- cancer_data %>% filter(treatment == "Cannabinoids") %>%  
  pull(tumor_volume)  
irr <- cancer_data %>% filter(treatment == "Irradiation") %>%  
  pull(tumor_volume)
```

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```
n_b <- t.test(neither, both)
c_b <- t.test(can, both)
i_b <- t.test(irr, both)
n_c <- t.test(neither, can)
n_i <- t.test(neither, irr)
c_i <- t.test(can, irr)
```

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Bonferroni for our example

and compare the p-value of each test to an alpha of 0.0083

```
p_values <- c(n_b$p.value, c_b$p.value, i_b$p.value,  
              n_c$p.value, n_i$p.value, c_i$p.value)  
names(p_values) <- c("n_b", "c_b", "i_b", "n_c", "n_i", "c_i")  
round(p_values, 4)
```

```
##      n_b      c_b      i_b      n_c      n_i      c_i  
## 0.0039 0.0539 0.0413 0.0921 0.7373 0.1411
```

What did Bonferroni conclude?

- ▶ Only one comparison clears the adjusted bar. Neither vs. Both, with $p = 0.0039$, which is below 0.0083
- ▶ Irradiation vs. Both has $p = 0.0413$. That would have been “significant” at 0.05 , but it is **not** significant once we correct for making 6 comparisons
- ▶ That is the price of the correction. We protect the overall type I error rate, and in exchange each individual test becomes harder to pass

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Tukey's HSD

Tukey's honest significant differences (Tukey's HSD)



John Wilder Tukey

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- ▶ Tukey's test maintains a 5% **experimentwise** or “**family**” error rate.
- ▶ Even if you make many pairwise comparisons, the overall error rate is 5% (at most)
- ▶ Overcomes the issue of multiple testing. Recall: If you conducted 100 tests with a 5% error rate (i.e., $\alpha = 0.05$) AND the H_0 was always true, how many p-values would you expect to be < 0.05 ?
- ▶ The Tukey's error rate is 5% **overall**, no matter how many tests you do. Thus it overcomes the problem of **multiple testing**

TukeyHSD() to calculate the differences in R

3+ Sample testing
with continuous
outcomes: ANOVA

You can think of the TukeyHSD() as a wrap-around for the anova, you can either nest the statements like this:

```
TukeyHSD(aov(outcome ~ group))
```

or save the ANOVA as an object and use that in the statement:

```
modelresult <- aov(outcome ~ group)
```

```
TukeyHSD(modelresult, conf.level = 0.95)
```

Note: `conf.level` is the **overall** confidence level for the whole family of comparisons, so it stays at 0.95.

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TukeyHSD() to calculate the differences in R

Here is the R code and output for the cancer example:

```
diffs <- TukeyHSD(cancer_anova, conf.level = 0.95) %>% tidy()
diffs %>% select(contrast, estimate, conf.low, conf.high, adj.p.value)
```

```
## # A tibble: 6 x 5
##   contrast      estimate conf.low conf.high adj.p.value
##   <chr>         <dbl>    <dbl>    <dbl>    <dbl>
## 1 Cannabinoids-Both      20.0    -14.3     54.3     0.378
## 2 Irradiation-Both      48.3     11.7     84.8     0.00756
## 3 Neither-Both          42.3     10.8     73.8     0.00661
## 4 Irradiation-Cannabinoids 28.3     -9.72    66.2     0.190
## 5 Neither-Cannabinoids    22.3    -10.9     55.4     0.263
## 6 Neither-Irradiation   -5.96    -41.4     29.5     0.964
```

TukeyHSD() to calculate the differences in R

Each row in the table corresponds to a pairwise test. The first row is the difference between Cannabinoids and Both. The estimated difference in means is 20, the interval runs from -14.3 to 54.3, and the adjusted p-value is 0.378.

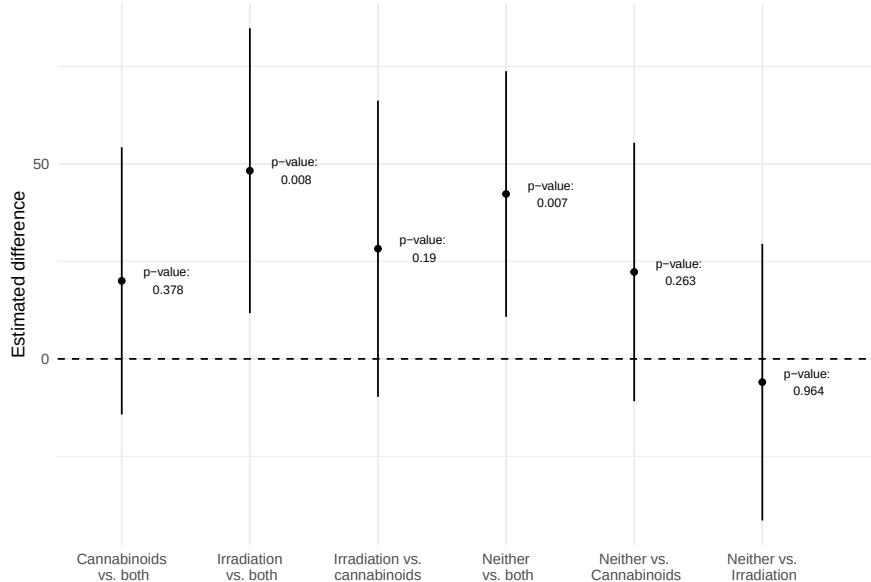
- ▶ “Adjusted” means adjusted for conducting multiple tests. The unadjusted p-value for that same pair was 0.054, so the adjustment made it substantially larger
- ▶ The interval Tukey reports is **also** adjusted, so it is a simultaneous 95% interval for the whole family. That is why it agrees with the adjusted p-value. It contains 0, and the adjusted p-value is above 0.05
- ▶ Read the interval and the p-value together. Both are telling you the same family-wise story

Which pairs does Tukey flag?

- ▶ Irradiation vs. Both, adjusted $p = 0.0076$
- ▶ Neither vs. Both, adjusted $p = 0.0066$
- ▶ Everything else has an adjusted p -value above 0.05

Tukey flags one more pair than Bonferroni did. Tukey is more powerful here because it pools all four groups into one estimate of the within-group variance with 18 degrees of freedom, while each separate t -test only had a handful.

Visualize the pairwise differences



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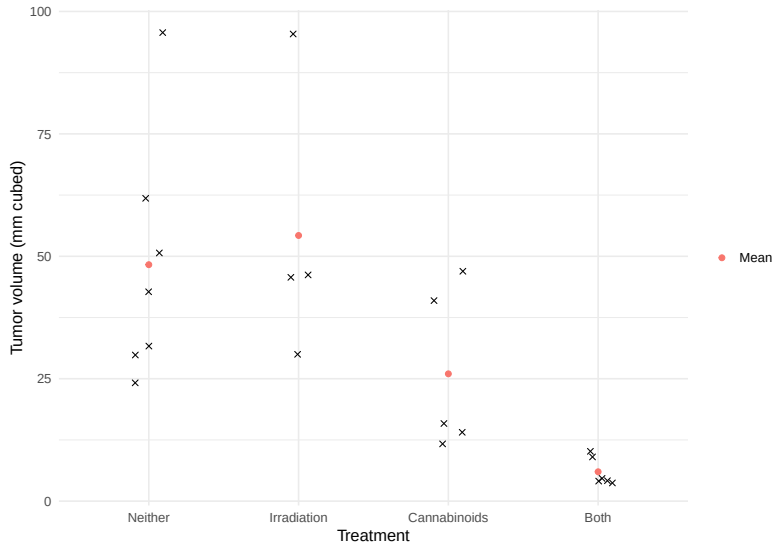
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Review raw data for comparison



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Putting it together

- ▶ The ANOVA said the four treatment means are not all equal, $F = 6.70$ on 3 and 18 degrees of freedom, $p = 0.003$
- ▶ The follow-up comparisons say where that came from. The combined treatment shrank tumors relative to irradiation alone and relative to no treatment at all
- ▶ We could not separate the combined treatment from cannabinoids alone. With 5 and 6 mice, that comparison is simply underpowered
- ▶ Remember the caveat. Condition 3 was violated, and these are mice, not people

Summary of today

- ▶ ANOVA tests one null hypothesis, that **all** the group means are equal, using one F statistic, so it avoids the multiple comparison problem the t-tests create
- ▶ F is a ratio, variation **between** the group means over variation **within** groups
- ▶ A significant ANOVA tells you the means are not all equal. It does **not** tell you which pair differs
- ▶ Bonferroni and Tukey are two ways to follow up. Bonferroni shrinks α , Tukey adjusts the p-values and the intervals directly
- ▶ Always look at the picture and check the conditions before you trust the p-value

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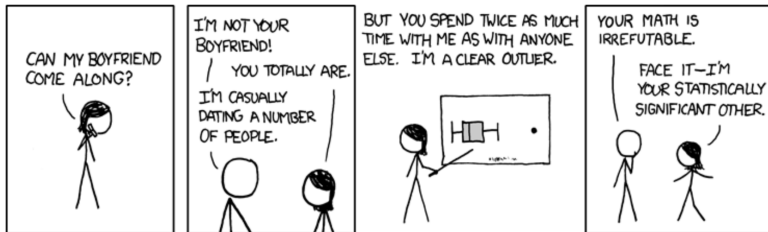
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